

Himanshu Kumar

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TECHNICAL SKILLS

Languages: Python, C++, Bash, SQL, HTML, CSS,

Frameworks/Libraries: Django, React.js, Node.js, OpenCV, Scikit-learn

Developer Tools: VS Code, Git, GitHub, Linux, Docker, Postman, GitHub Actions

Databases: MongoDB, MySQL

Core Concepts: Data Structures, Algorithms, Operating Systems, Computer Networks, Distributed Systems, Parallel Computing (MPI, OpenMP, CUDA), High-Performance Computing (HPC), REST APIs, CI/CD, Performance Optimization

EXPERIENCE

HPC Intern

Aug 2025 – Feb 2026

C-DAC

Kolkata

- Developed and optimized high-performance distributed systems for large-scale simulations and data processing.
- Implemented parallel computing using MPI and OpenMP, reducing execution time by 40%.
- Worked on performance optimization, concurrency, and efficient resource utilization.

Software Developer

Dec 2023 – July 2025

SMS Techsoft

Bangalore

- Developed and maintained full-stack web applications using Python, JavaScript, SQL, HTML/CSS, and REST APIs, contributing to scalable client-based business solutions.
- Debugged production issues, performed functional and application testing, and optimized backend/frontend components to improve system reliability, usability, and performance.
- Collaborated in Git-based development workflows, deployment/testing cycles, requirement analysis, and feature enhancement across multiple projects in a team-oriented environment.

Full Stack Developer Intern

Aug 2023 – Nov 2023

NewArk India Pvt Ltd.

Aurangabad

- Developed scalable web applications leveraging the **MERN stack**, delivering **RESTful APIs** that handled over 10,000 daily requests and implemented robust user authentication for 5,000+ active users.
- Collaborated with a cross-functional team of 10 members to gather requirements, translating them into technical specifications, and successfully deploying applications on Platforms, reducing deployment time by 20%.
- Developed an AI-powered chatbot to assist students and website visitors with seamless access to information, leveraging Hugging Face models and BeautifulSoup for data processing and retrieval.

PROJECTS

SmartTraffic-HPC | C++, OpenCV, YOLOv8, MPI, OpenMP, CUDA

[source code](#)

- Built a high-performance traffic detection system using YOLOv8 and OpenCV for real-time vehicle analysis.
- Achieved 200+ FPS throughput using parallel computing techniques (MPI, OpenMP, CUDA).
- Designed a hybrid CPU-GPU pipeline for scalable and efficient real-time processing.
- Deployed solution on HPC clusters for large-scale traffic analytics and simulation.

Rice Grain Analysis | Python, OpenCV, Scikit-learn

[source code](#)

- Developed a machine learning system to classify rice grains based on image features.
- Built ML-based image classification system using OpenCV and Scikit-learn for agricultural quality analysis.
- Applied image processing techniques for feature extraction including size, shape, and texture.

EDUCATION

Magadh University

BSC in Information Technology – 72.80%

Bodh Gaya, Bihar

August 2020 – March 2024

ACHIEVEMENTS

- Qualified GATE 2026 (Computer Science) – AIR 6123 (Score: 524)